Facultad de Ciencias
UNAM

AVANCES DE TESIS DE LICENCIATURA

Cálculo numérico de propiedades ópticas de eritrocitos sanos y enfermos

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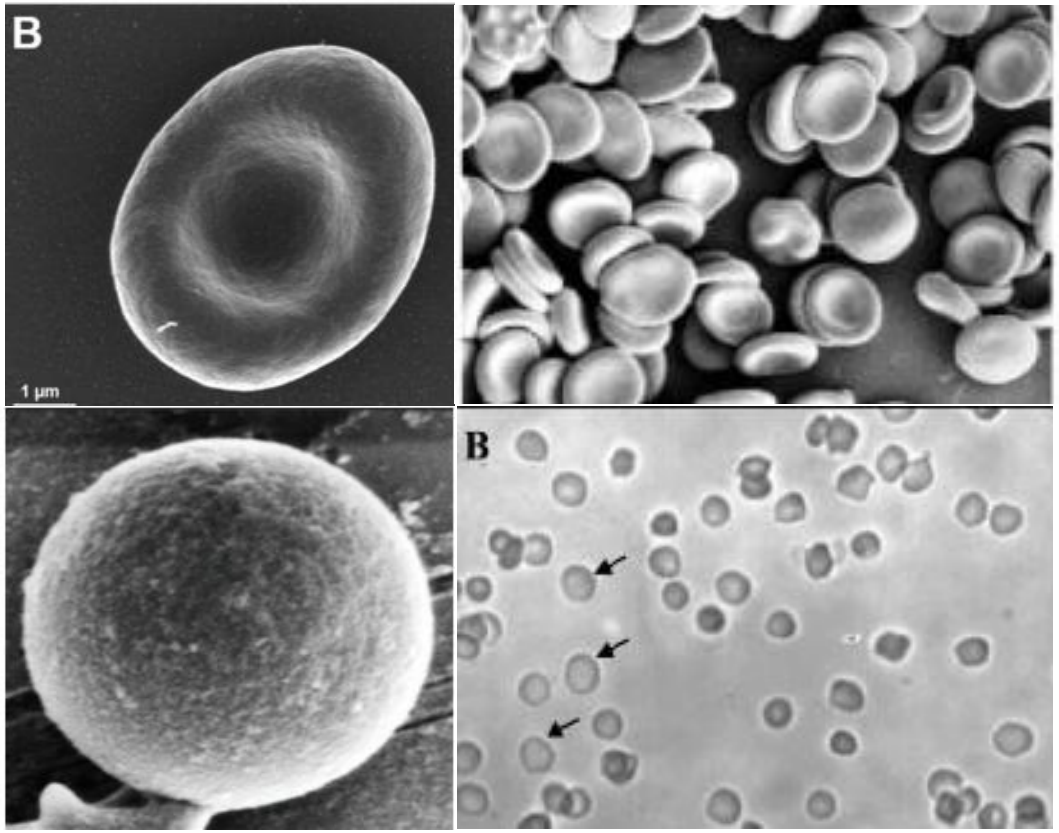
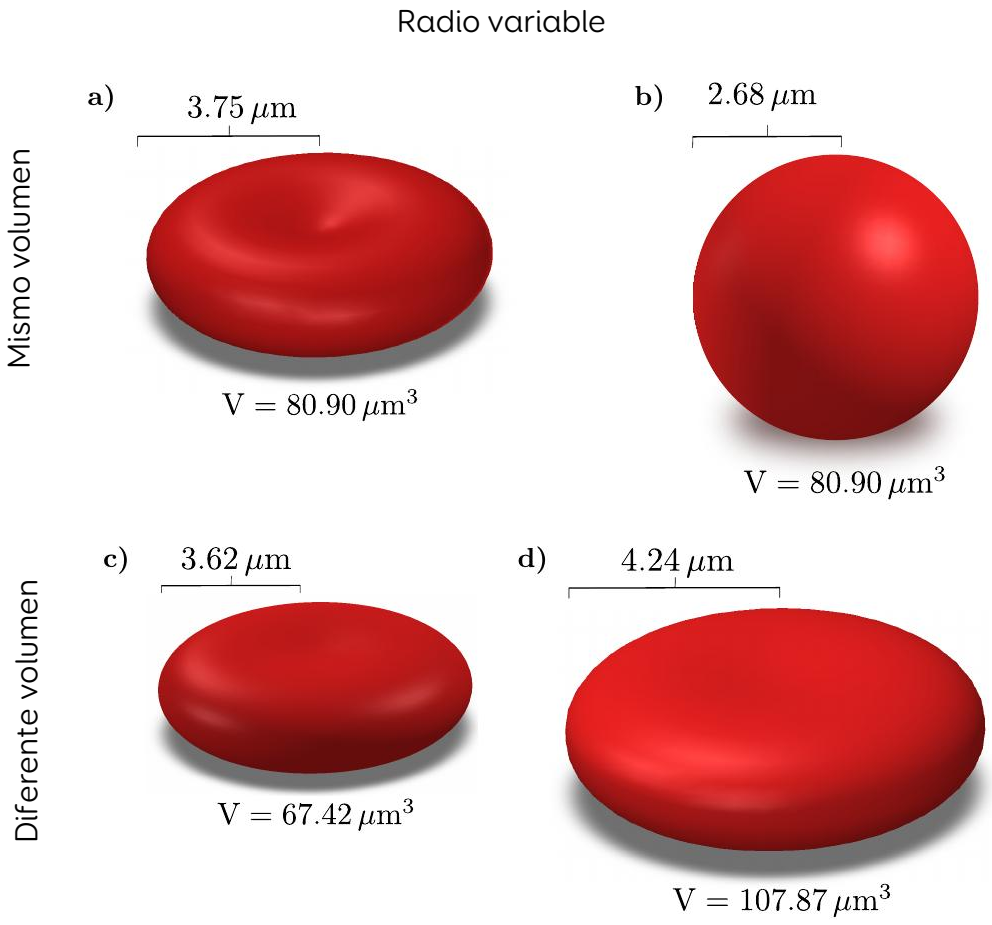
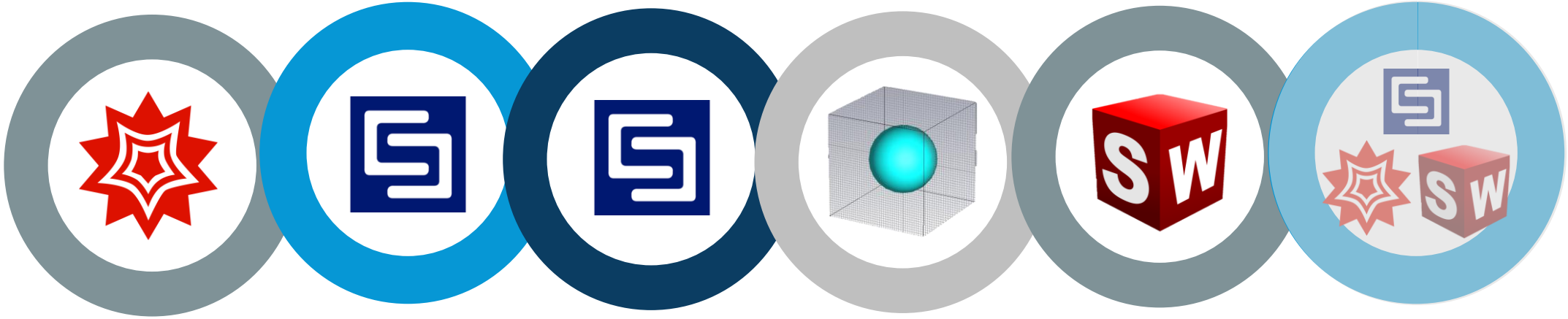
Antecedentes





Alummoottil, M. , *et al.* *Hemato*, **15**(4), 2010
Khanna, R., *et al.* *Proceedings of the National Academy of Sciences*, **99**(10), 2002

Plan de trabajo



- Sano
- Microcitosis
- Macrocitosis
- Esferocitosis hereditaria

Construcción de funciones dieléctricas de eritrocitos

Comparación de CST vs Mie

Análisis de convergencia numérica en CST

Evaluación del efecto del mallado

Construcción de geometrías de eritrocitos

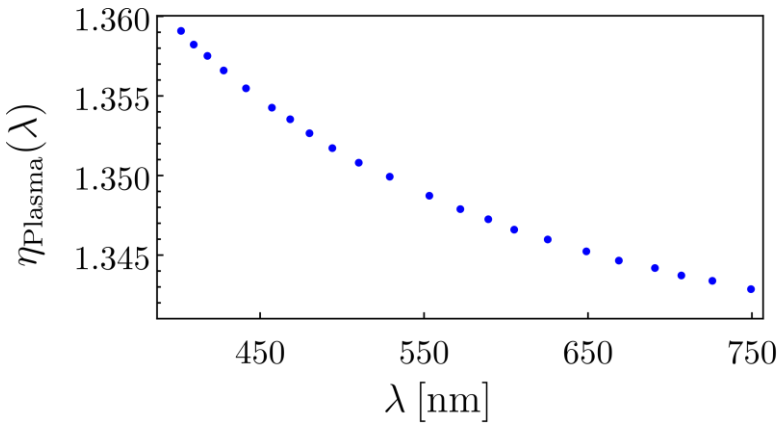
Cálculo de propiedades ópticas de eritrocitos

Redacción → Teoría → Resultados y conclusiones

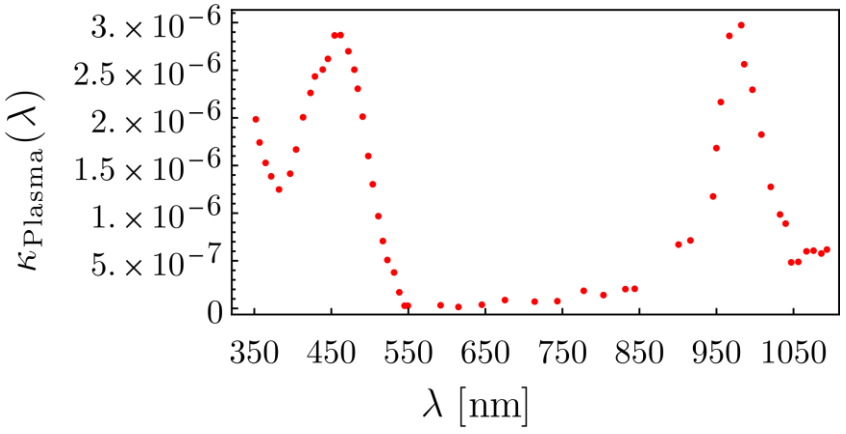
Datos experimentales

$$n(\lambda) = \eta(\lambda) + i\kappa(\lambda)$$

Plasma



Liu, S., *et al.* *Journal of Biomedical Optics*, **24**(3), 2019



Meinke, M., *et al.* *Journal of Biomedical Optics*, **12**(1), 2007

Eritrocitos

Parte real:

Rango espectral

250 - 1100 nm

$$\eta_{\text{Hb}}(\lambda, c_{\text{Hb}}) = \eta_{\text{H}_2\text{O}}(\lambda)[\beta(\lambda)c_{\text{Hb}} + 1]$$

Friebel, M. y Meinke, M. *Applied Optics*, **45**(12), 2006

Parte imaginaria:

	κ_{Hb}	Rango espectral
• 28.7 g/dl	Microcitosis	250 - 600 nm
• 30.6 g/dL	Sano	250 - 1050 nm
• 34 g/dL	Macrocitosis/ esferocitosis	250 - 600 nm

Friebel, M. y Meinke, M. *Applied Optics*, **10**(6), 2005

Funciones dieléctricas

1. Obtención de la parte real e imaginaria de la función dieléctrica mediante las relaciones

$$\varepsilon' = \eta^2 - \kappa^2$$
$$\varepsilon'' = 2\eta\kappa$$

2. Ajuste de lorentzianas a la parte imaginaria de la función dieléctrica

$$\varepsilon'' = \sum_{i=1}^{5} \frac{A_i \gamma_i \omega}{(\omega_{0_i}^2 - \omega^2)^2 + \gamma_i^2 \omega^2}$$

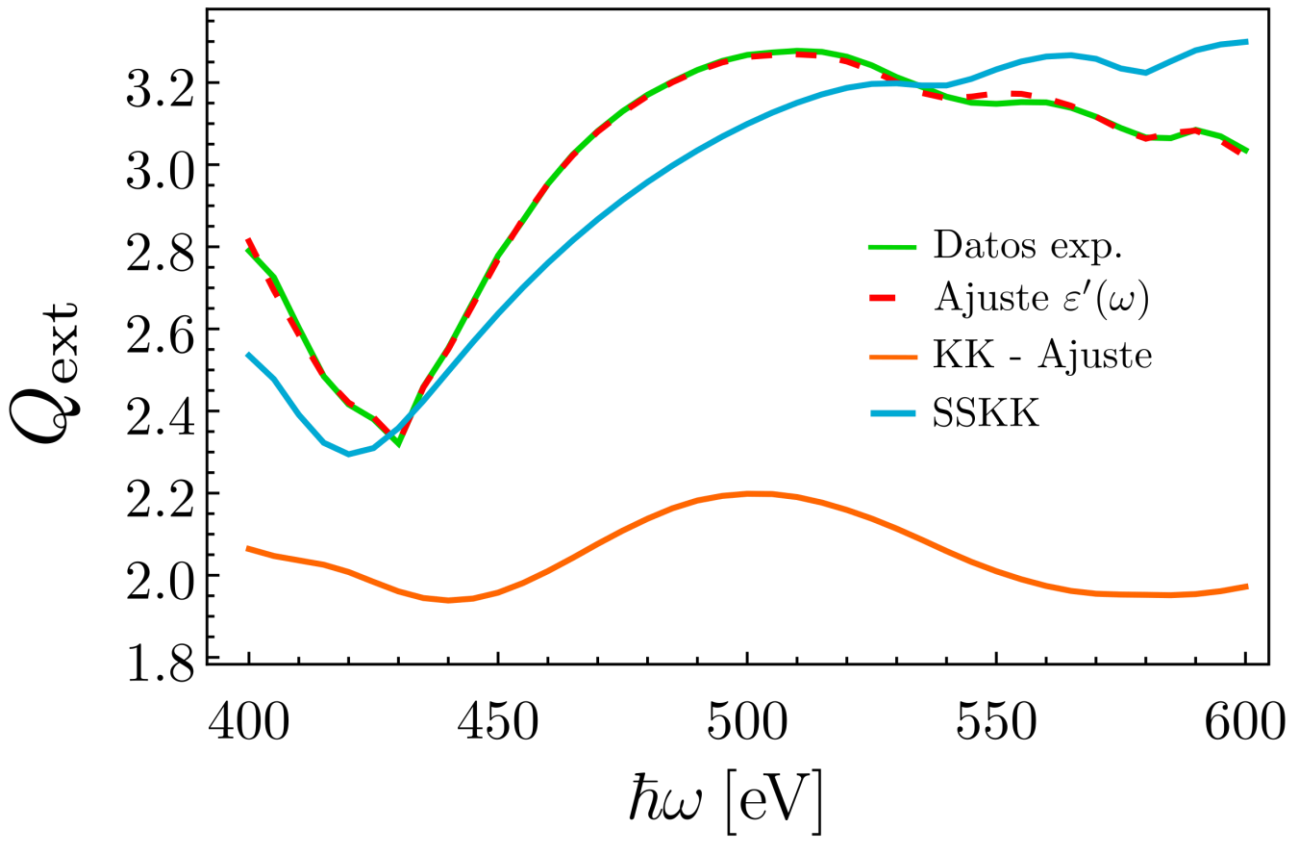
3. Obtención de la parte real e imaginaria de la función dieléctrica

$$\varepsilon'(\omega) = \varepsilon'(\omega_1) + \frac{2(\omega^2 - \omega_1^2)}{\pi} \mathrm{PV} \int_0^\infty \frac{\omega' \varepsilon''(\omega')}{(\omega'^2 - \omega^2)(\omega'^2 - \omega_1^2)} \mathrm{d}\omega'$$
$$\varepsilon''(\omega) = \varepsilon''(\omega_1) - \frac{2}{\pi}(\omega - \omega_1) \mathrm{PV} \int_0^\infty \frac{(\varepsilon'(\omega') - 1)(\omega'^2 + \omega\omega_1)}{(\omega'^2 - \omega^2)(\omega'^2 - \omega_1^2)} \mathrm{d}\omega'$$

Eficiencia de extinción

Teoría de Mie

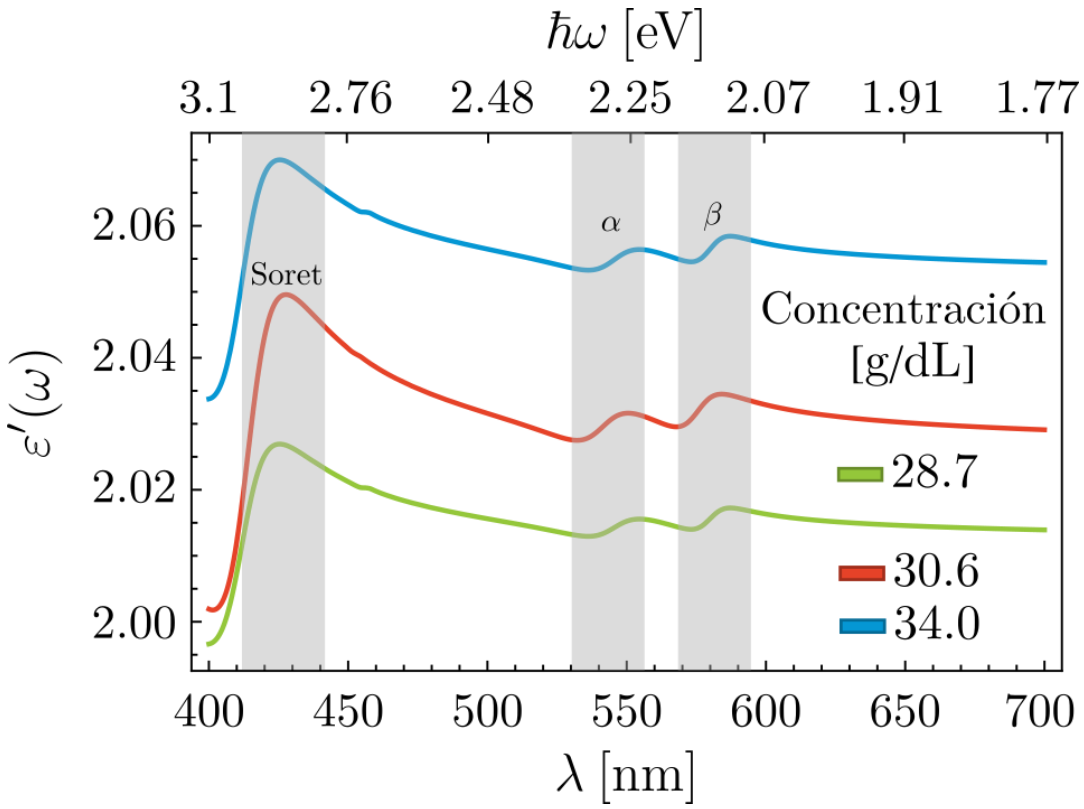
Q_{ext} empleando diferentes funciones dieléctricas de la hemoglobina

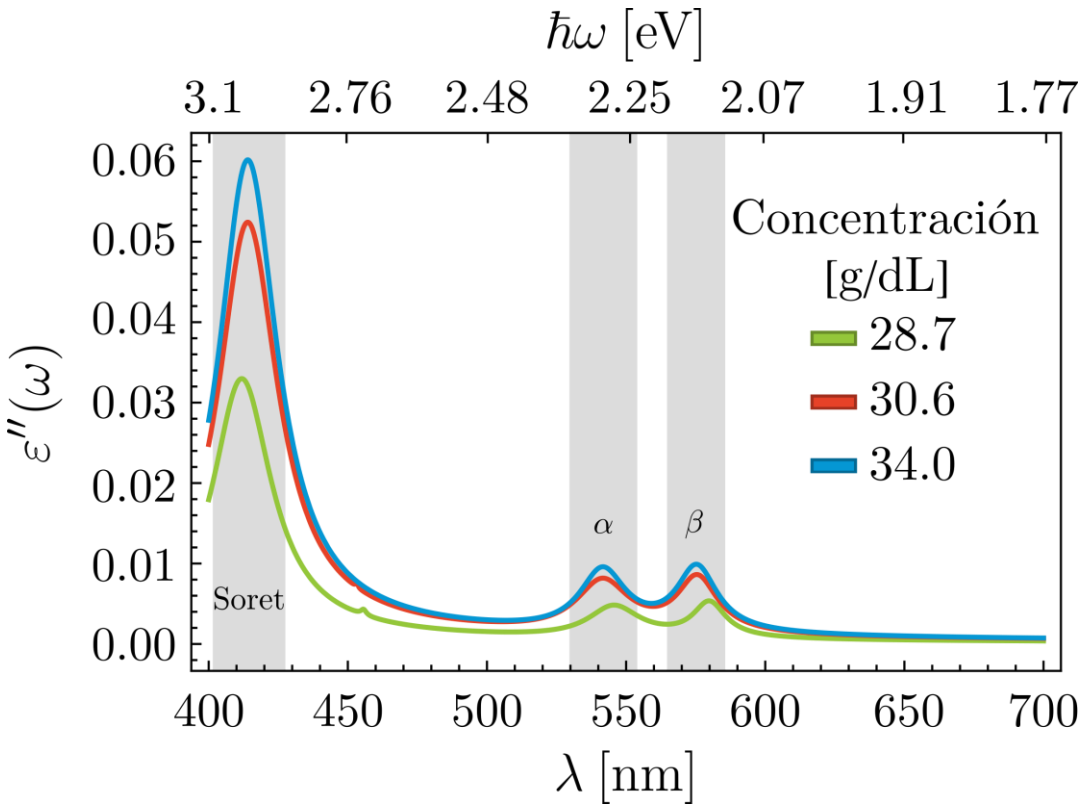


Radio: 2680 nm

Matriz: Plasma ($n_{\text{Plasma}} = 3.31$)

Función dieléctrica

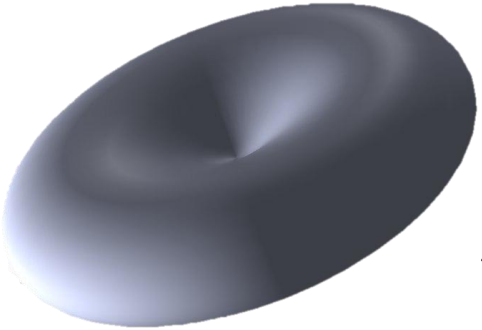




SolidWorks

Eritrocitos

Eritrocito sano



$$a = 2.6\,\mu\text{m}$$
$$b = 2.7\,\mu\text{m}$$
$$c = 0.8\,\mu\text{m}$$
$$\text{Radio} = 3.76\,\mu\text{m}$$
$$\text{Volumen} = 79.2\,\mu\text{m}^3$$

Microcito

$$a = 2.4\,\mu\text{m}$$
$$b = 2.7\,\mu\text{m}$$
$$c = 0.5\,\mu\text{m}$$
$$\text{Radio} = 3.62\,\mu\text{m}$$
$$\text{Volumen} = 70.40\,\mu\text{m}^3$$

Macrocito

$$a = 2.8\,\mu\text{m}$$
$$b = 3.2\,\mu\text{m}$$
$$c = 0.5\,\mu\text{m}$$
$$\text{Radio} = 4.25\,\mu\text{m}$$
$$\text{Volumen} = 104.51\,\mu\text{m}^3$$

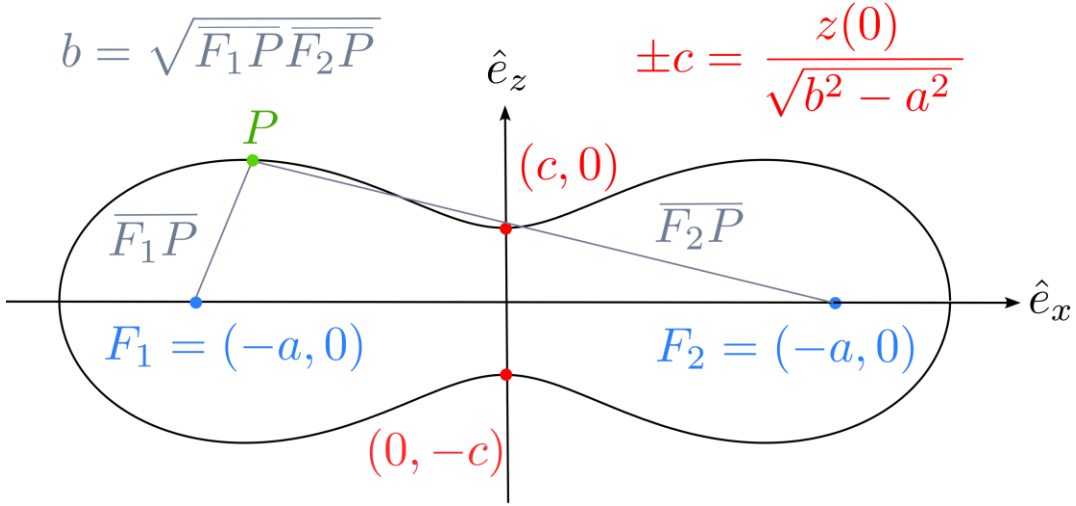
Esferocito

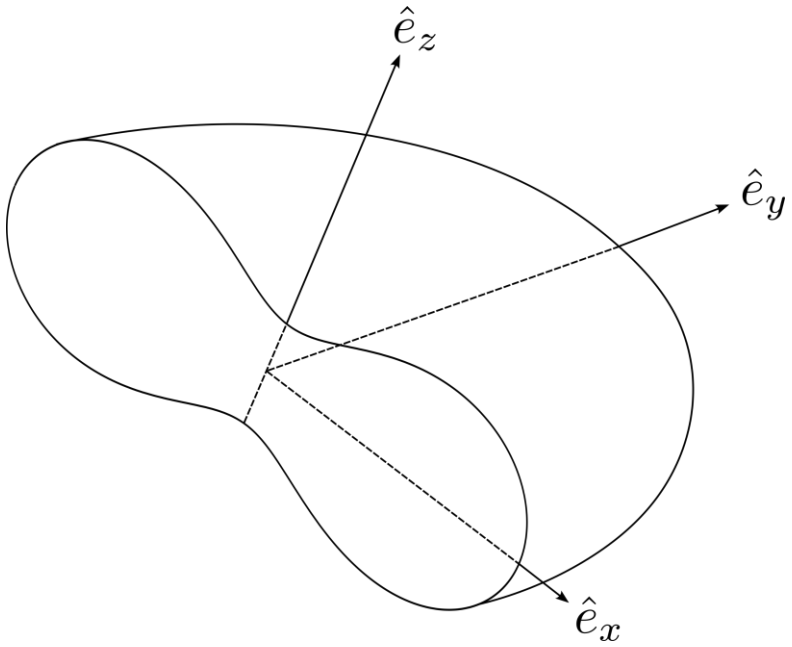
$$\text{Radio} = 2.68\,\mu\text{m}$$
$$\text{Volumen} = 80.63\,\mu\text{m}^3$$

J. Q. Lu, et al. *Journal of Biomedical Optics*, **10**(2), 2005

Óvalos de Cassini

$$z = \pm c(-a^2 - x^2 \pm (4x^2a^2 + b^4)^{1/2})^{1/2}$$

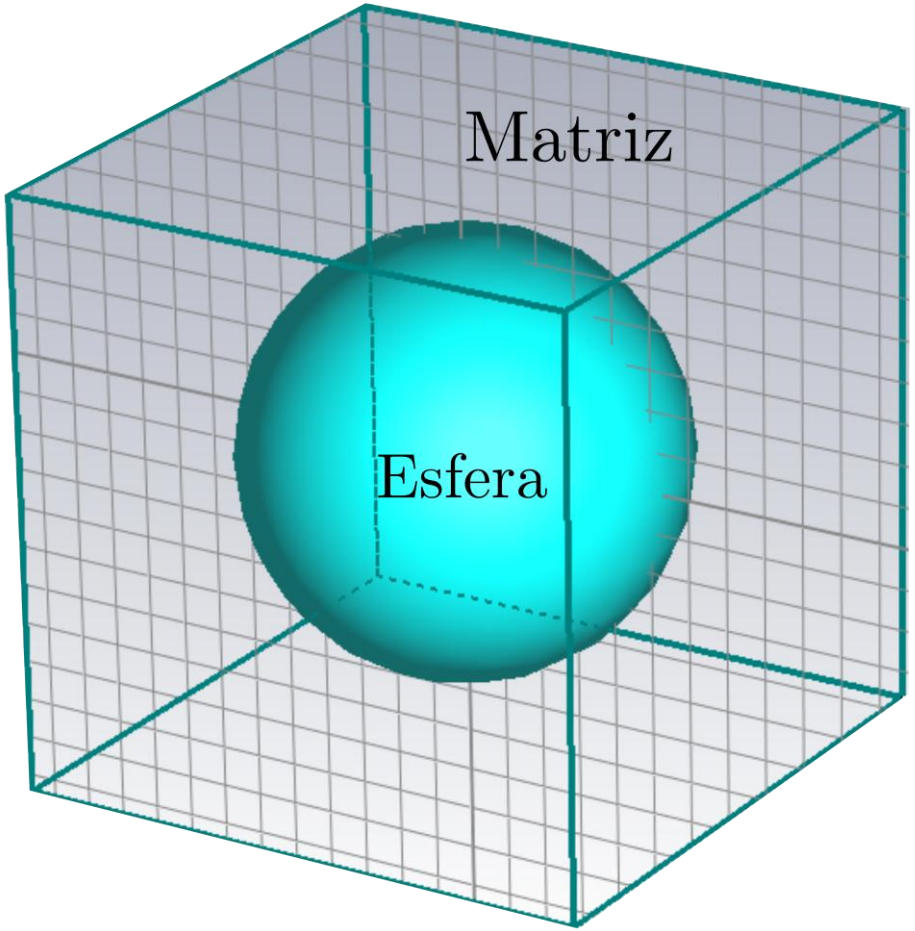




J. Q. Lu, et al. *Journal of Biomedical Optics*, **10**(2), 2005

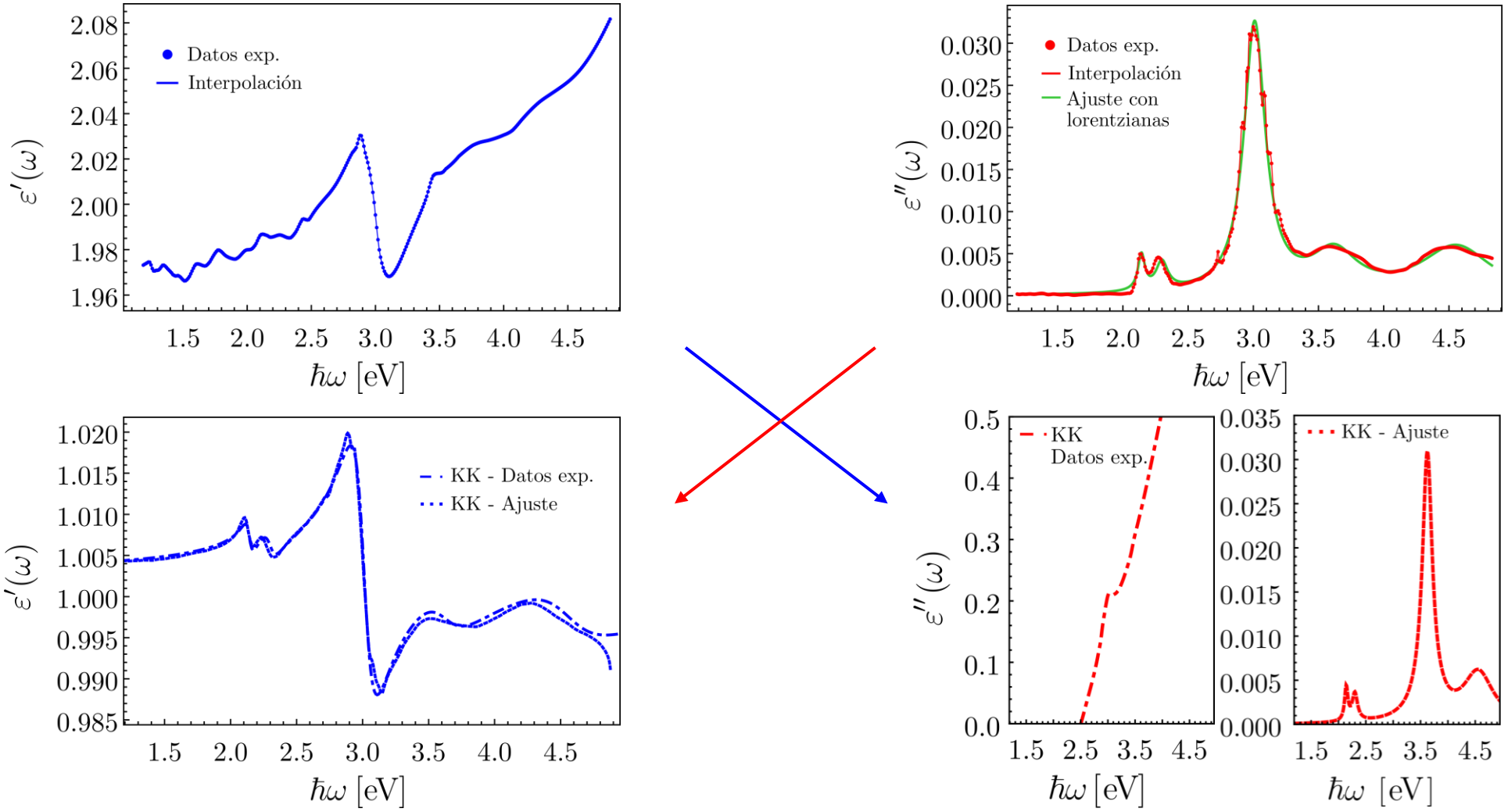
CST Studio vs Mie

Convergencia



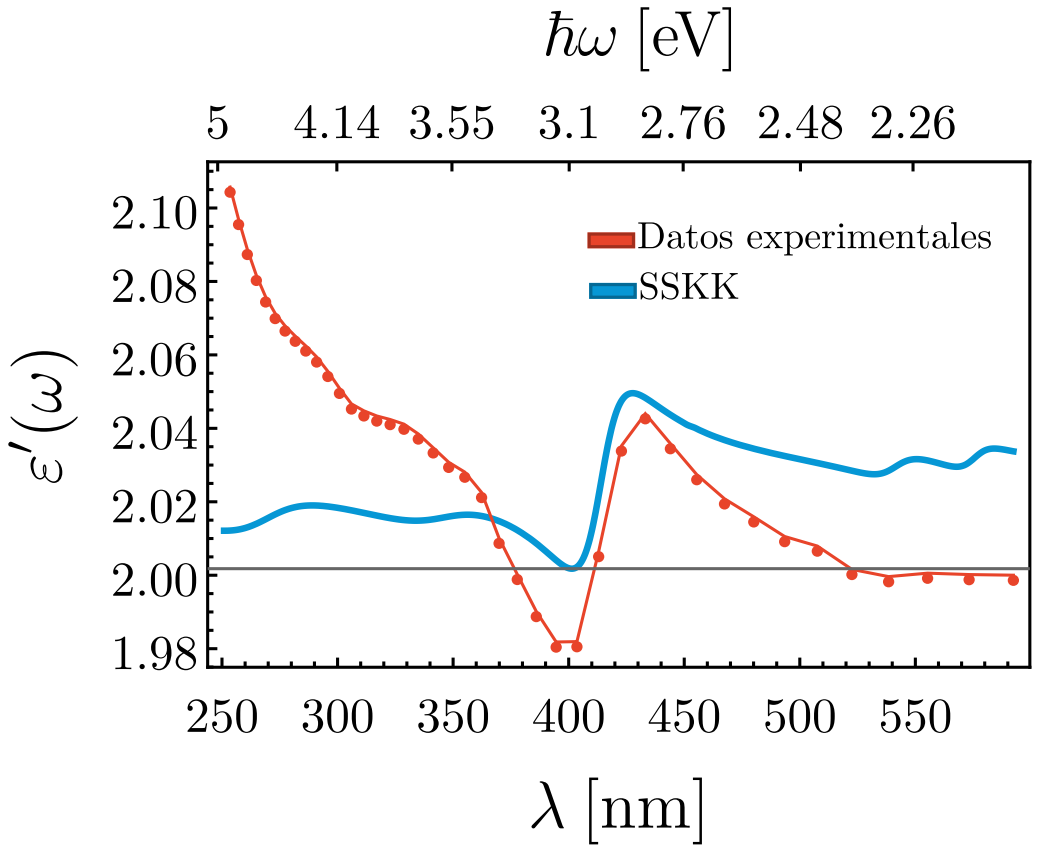
Mallado

Relaciones de KK



Friebel, M., y Meinke, M. *Applied Optics*, **10**(6), 2005

Relaciones de SSKK



Friebel, M., y Meinke, M. *Applied Optics*, **10**(6), 2005